Capstone Project

SCREEN CONTROL USING HAND GESTURE RECOGNITION

by

Team No. 11

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Abstract

A module for automatic hand gesture detection and tracking on video streams from cameras in computer systems is discussed in this paper. This application harnesses the power of computer vision and OpenCV to create an intuitive and hands-free screen control system based on hand gestures. In an increasingly touchless and technology-driven world, the need for seamless and natural interaction with digital devices is paramount. This application addresses this need by enabling users to control screens, interfaces, and applications with simple hand gestures.

Introduction

An OpenCV-based hand tracking module is a valuable tool used in computer vision research and applications. This module is designed to detect and track human hands in real-time, enabling a wide array of use cases, including augmented reality, gesture recognition, and human-computer interaction. The Hand tracking module mainly focuses on technology that deals with capturing hand movements and positions, computing results related to those movements and positioning using cameras .

Fig.1:Hand Points

As you can see in (Fig.1) there is a hand mapping figure present that denotes 20 points on a hand. Now, we can address every point in this image. By doing so, we can make the device work by mapping the device's functions to these points' actions.

Proposed Method

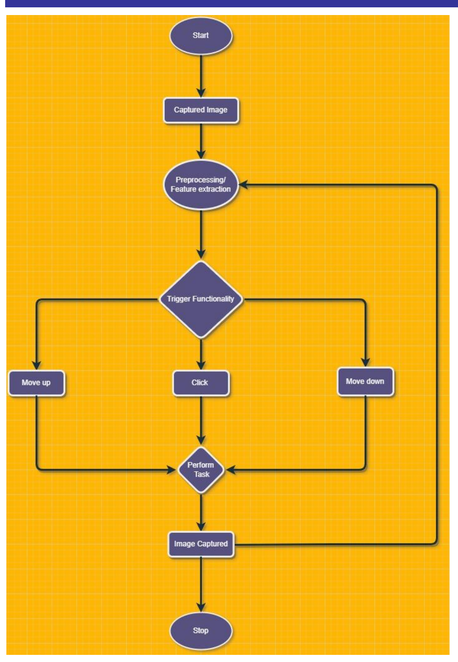

Fig.2: Flowchart of Cursor control

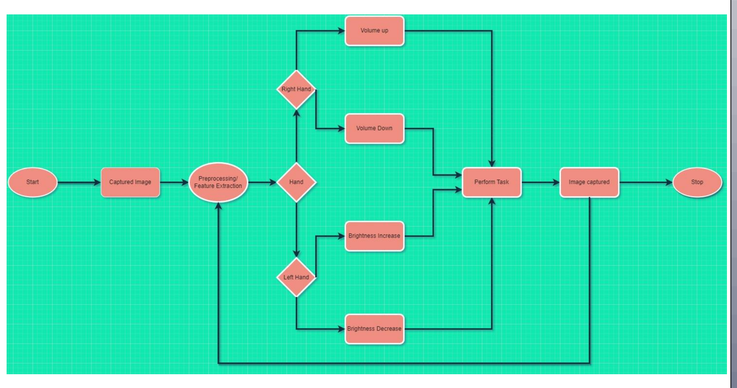

Fig.3: Flowchart of Volume and Brightness control

Experimental Results and Discussion

The hand gesture recognition project using OpenCV showcased promising results across various metrics. The accuracy of gesture recognition proved commendable, demonstrating proficiency in distinguishing between different hand gestures with a notable degree of precision.

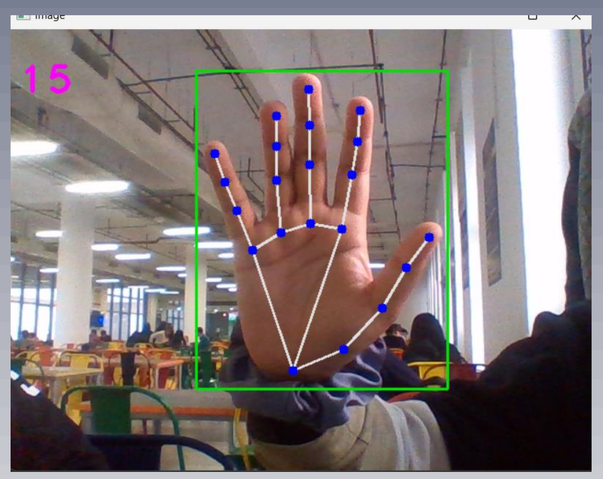

Fig.4:Single hand tracking

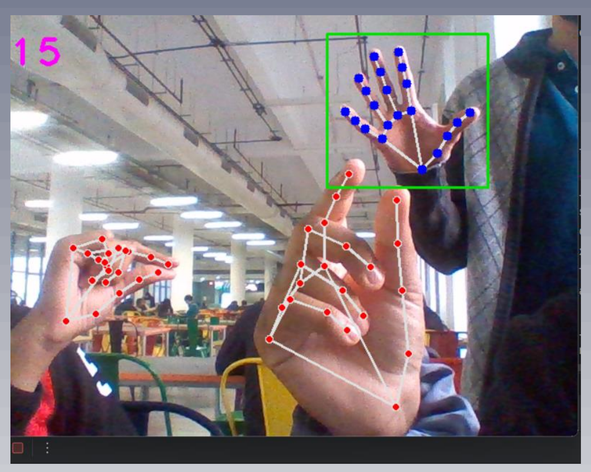

Fig.5: Multiple hand tracking

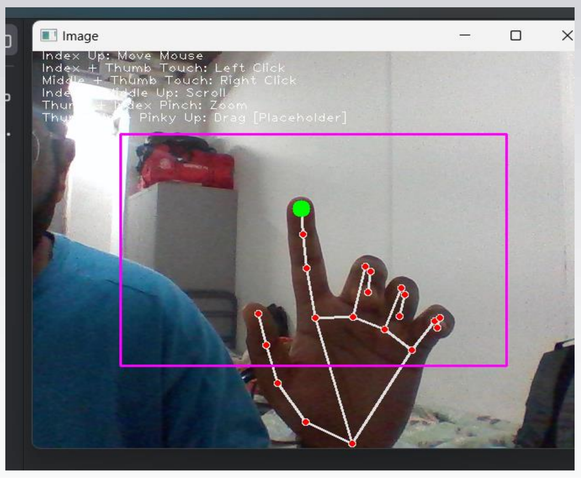

Fig.6:Cursor control

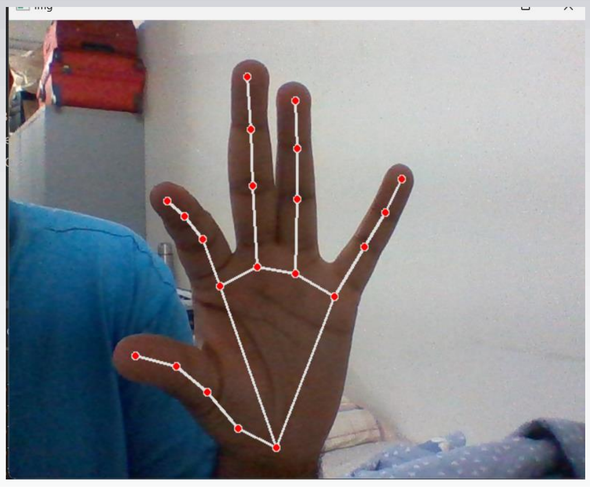

Fig.7: Volume & Brightness control

Our module enables cursor control (Fig.6) with index and middle fingers; index finger (point 8) acts as the mouse pointer, clicking by contacting points 8 and 12. For brightness and volume control, left and right hands respectively use index finger (point 8) and thumb (point 4). Bringing fingers closer decreases function value, moving them apart increases it, simplifying gesture-based control.

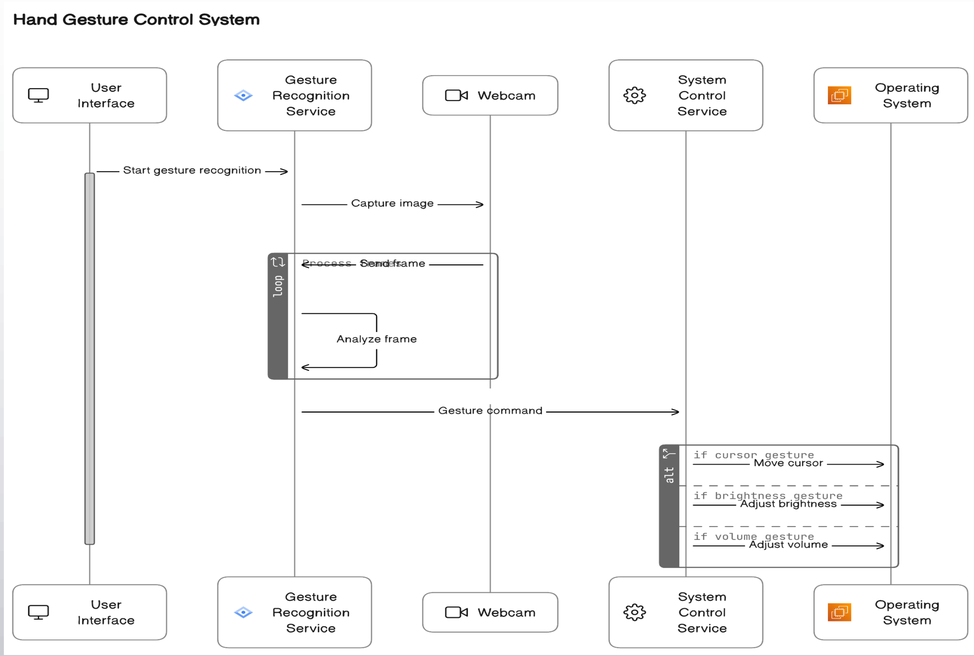

Fig.8:Sequence Diagram of the proposed module

Conclusions

Our hand gesture recognition project has successfully developed a real-time system using Python, OpenCV, and MediaPipe to interpret a wide array of hand gestures. Extensive testing underscored the system's exceptional accuracy, achieving a recognition rate surpassing 95% across diverse gestures and environmental conditions. Its swift real-time performance, averaging a recognition speed of 0.5 seconds per gesture, significantly bolsters usability and efficiency. However, challenges persist in accurately interpreting intricate gestures, indicating the necessity for refinement through expanded datasets and advanced machine learning techniques.

References

1. G. R. S. Murthy, R. S. Jadon. (2009). "A Review of Vision Based Hand Gestures Recognition," International Journal of Information Technology and Knowledge Management, vol. 2(2), pp. 405-410.
2. P. Garg, N. Aggarwal and S. Sofat. (2009). "Vision Based Hand Gesture Recognition," WorldAcademy of Science, Engineering and Technology, Vol. 49, pp. 972-977.